PART 1 – PHYSICAL & CHEMICAL CHANGES

Identification

Directions: Identify the term, concept, process, reaction type, or symbol described in each statement. Write your answer on the blank.

1. A change that produces one or more new substances.
   Answer: ______________________
2. A change that only affects the size, shape, or state of a substance without changing its composition.
   Answer: ______________________
3. The process of tomatoes spoiling and producing new substances.
   Answer: ______________________
4. The type of change represented by crushing chunks of ice.
   Answer: ______________________
5. The main characteristic that distinguishes a chemical change from a physical change.
   Answer: ______________________
6. The evidence of a chemical reaction shown when vinegar reacts with baking soda and bubbles are produced.
   Answer: ______________________
7. The evidence of a chemical reaction shown when milk forms lumpy solids.
   Answer: ______________________
8. The evidence shown when quicklime reacts with water and releases heat.
   Answer: ______________________
9. The change that occurs when paper is cut into smaller pieces.
   Answer: ______________________
10. The evidence of a chemical reaction when a yellow solid forms after mixing two clear solutions.
    Answer: ______________________
11. The substances found on the left side of a chemical equation.
    Answer: ______________________
12. The substances formed on the right side of a chemical equation.
    Answer: ______________________
13. The meaning of the symbol **(aq)** in a chemical equation.
    Answer: ______________________
14. The meaning of the symbol **Δ** written above the reaction arrow.
    Answer: ______________________
15. The meaning of the downward arrow (↓) in a chemical equation.
    Answer: ______________________
16. The meaning of the plus sign (+) between substances in a chemical equation.
    Answer: ______________________

17. The reaction type represented by **$A + B \rightarrow AB$**.
    Answer: ______________________
18. The reaction type shown by **$2Na + Cl_2 \rightarrow 2NaCl$**.
    Answer: ______________________
19. The reaction type represented by **$CaCO_3 \rightarrow CaO + CO_2$**.
    Answer: ______________________
20. The general equation for a decomposition reaction.
    Answer: ______________________
21. The reaction in which one element replaces another element in a compound.
    Answer: ______________________
22. The reaction type represented by **$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$**.
    Answer: ______________________
23. The gas that is always required for combustion reactions.
    Answer: ______________________
24. The two products always formed during the complete combustion of a hydrocarbon.
    Answer: ______________________
25. The reaction represented by **$HBr + NaOH \rightarrow NaBr + H_2O$**.
    Answer: ______________________
26. The special type of double displacement reaction involving an acid and a base.
    Answer: ______________________
27. The type of change represented by melting wax.
    Answer: ______________________
28. The process represented by dissolving salt in water.
    Answer: ______________________
29. In the equation **$Mg + O_2 \rightarrow MgO$**, the substance **MgO** is called the ________.
    Answer: ______________________
30. The meaning of the reaction arrow ($\rightarrow$) in a chemical equation.
    Answer: ______________________

Answer Key

1. Chemical change
2. Physical change
3. Rotting (rotting of tomatoes)
4. Physical change
5. Formation of a new substance
6. Gas evolution (gas formation)
7. Formation of a precipitate
8. Temperature change (heat released)
9. Physical change
10. Precipitate formation
11. Reactants
12. Products

13. Aqueous (dissolved in water)
14. Heat is supplied
15. Precipitate (solid formed)
16. Reacts with / mixed with
17. Combination (Synthesis) reaction
18. Combination (Synthesis) reaction
19. Decomposition reaction
20. **$AB \rightarrow A + B$**
21. Single displacement reaction
22. Double displacement reaction
23. Oxygen (O_2)
24. Carbon dioxide (CO_2) and water (H_2O)
25. Neutralization (Acid–Base reaction)
26. Neutralization
27. Physical change
28. Physical change
29. Product
30. Yields / Produces

PART 2 – ACIDS, BASES, AND SALTS

Identification

Directions: Identify the term, concept, compound, indicator, ion, or process described in each statement. Write your answer on the blank.

1. A substance that has a sour taste and turns blue litmus paper red.
   Answer: ______________________
2. A substance that turns red litmus paper blue and has a pH greater than 7.
   Answer: ______________________
3. The pH value of pure water at 25°C.
   Answer: ______________________
4. The color of phenolphthalein in a basic solution.
   Answer: ______________________
5. The ion released by acids in water according to the Arrhenius definition.
   Answer: ______________________
6. The ion released by bases in water according to the Arrhenius definition.
   Answer: ______________________
7. The chemical formula of common table salt.
   Answer: ______________________
8. The chemical formula of hydrochloric acid found in the stomach.
   Answer: ______________________
9. The chemical formula of sodium hydroxide (caustic soda).
   Answer: ______________________

10. The color of methyl orange in an acidic solution.
    Answer: ______________________
11. A natural indicator obtained from plants that turns reddish-brown in a base.
    Answer: ______________________
12. A solution with a pH of 7 is described as __________.
    Answer: ______________________
13. The weak acid commonly found in vinegar.
    Answer: ______________________
14. The base commonly found in milk of magnesia.
    Answer: ______________________
15. The color of blue litmus paper when placed in a base.
    Answer: ______________________
16. The reaction between an acid and a base that produces salt and water.
    Answer: ______________________
17. The type of indicator that changes smell instead of color in acids and bases.
    Answer: ______________________
18. The approximate mouth pH at which tooth enamel begins to dissolve.
    Answer: ______________________
19. The salt that forms a basic solution when dissolved in water.
    Answer: ______________________
20. The salt that forms an acidic solution when dissolved in water.
    Answer: ______________________
21. The reason dry hydrogen chloride gas does not affect dry blue litmus paper.
    Answer: ______________________
22. The factor by which hydrogen ion concentration changes for every one-unit change in pH.
    Answer: ______________________
23. Arrange the following pH values from strongest acid to strongest base: 2, 6, 7, 9, 12.
    Answer: ______________________
24. The neutral salt produced by the reaction of hydrochloric acid and sodium hydroxide.
    Answer: ______________________
25. The particles that allow acidic and basic solutions to conduct electricity.
    Answer: ______________________
26. The safe laboratory practice when diluting concentrated acids.
    Answer: ______________________
27. The gas released when baking soda reacts with an acid.
    Answer: ______________________
28. The indicator that remains blue in a basic solution.
    Answer: ______________________
29. The substance released by bases that gives them their basic properties.
    Answer: ______________________
30. The product formed together with water during a neutralization reaction.
    Answer: ______________________

Answer Key

1. Acid
2. Base
3. pH 7
4. Bright pink
5. H^+ (Hydrogen ion) / H_3O^+
6. OH^- (Hydroxide ion)
7. NaCl
8. HCl
9. NaOH
10. Red
11. Turmeric
12. Neutral
13. Acetic acid (CH_3COOH)
14. Magnesium hydroxide [$Mg(OH)_2$]
15. Blue
16. Neutralization
17. Olfactory indicator
18. pH 5.5
19. Sodium carbonate (Na_2CO_3)
20. Ammonium chloride (NH_4Cl)
21. No water is present to ionize HCl and produce H^+ ions.
22. 10×
23. $2 \rightarrow 6 \rightarrow 7 \rightarrow 9 \rightarrow 12$
24. Sodium chloride (NaCl)
25. Mobile ions
26. Pour acid slowly into water while stirring.
27. Carbon dioxide (CO_2)
28. Blue litmus paper
29. Hydroxide ion (OH^-)
30. Salt

PART 3 – TYPES OF CHEMICAL REACTIONS IN THE ENVIRONMENT

Identification

Directions: Identify the type of chemical reaction, process, or concept described in each statement. Write your answer on the blank.

1. The reaction represented by $\mathbf{A + B \rightarrow AB}$.
   Answer: ______________________

2. The reaction represented by **AB → A + B**.
   Answer: ______________________
3. The reaction in which a hydrocarbon reacts with oxygen to produce carbon dioxide, water, and heat.
   Answer: ______________________
4. The reaction between an acid and a base that forms salt and water.
   Answer: ______________________
5. The reaction represented by **AB + C → AC + B**.
   Answer: ______________________
6. The reaction represented by **AB + CD → AD + CB**.
   Answer: ______________________
7. The reaction type shown by $\mathbf{2Na + Cl_2 \rightarrow 2NaCl}$.
   Answer: ______________________
8. The reaction type represented by $\mathbf{CaCO_3 \rightarrow CaO + CO_2}$.
   Answer: ______________________
9. The process by which plants produce glucose from carbon dioxide and water using sunlight.
   Answer: ______________________
10. The process by which body cells break down glucose to release energy.
    Answer: ______________________
11. A reaction in which one reactant breaks down into two or more simpler substances.
    Answer: ______________________
12. The slow oxidation of iron that produces rust.
    Answer: ______________________
13. A reaction that forms an insoluble solid from two solutions.
    Answer: ______________________
14. A reaction in which one element replaces another element in a compound.
    Answer: ______________________
15. The biological process that is the reverse of photosynthesis.
    Answer: ______________________
16. The reaction represented by $\mathbf{Zn + 2HCl \rightarrow ZnCl_2 + H_2}$.
    Answer: ______________________
17. The reaction represented by $\mathbf{AgNO_3 + NaCl \rightarrow AgCl + NaNO_3}$.
    Answer: ______________________
18. The combustion of methane represented by $\mathbf{CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + heat}$.
    Answer: ______________________
19. The environmental problem caused when sulfur dioxide and nitrogen oxides react with rainwater.
    Answer: ______________________
20. The process represented by $\mathbf{4Fe + 3O_2 + H_2O \rightarrow 2Fe_2O_3 \cdot H_2O}$.
    Answer: ______________________
21. The process that is considered the opposite of cellular respiration.
    Answer: ______________________
22. The reaction that occurs when acid rain reacts with limestone statues.
    Answer: ______________________

23. The greenhouse gas commonly released during the burning of fuels, forests, and volcanic eruptions.
    Answer: ______________________
24. The process by which plants help reduce global warming by removing carbon dioxide from the atmosphere.
    Answer: ______________________
25. A reaction that has only one reactant and produces two or more products.
    Answer: ______________________
26. The reaction type represented by $\mathbf{2Mg + O_2 \rightarrow 2MgO}$.
    Answer: ______________________
27. The reaction type that always produces heat according to the source material.
    Answer: ______________________
28. The reaction represented by $\mathbf{2H_2O \rightarrow 2H_2 + O_2}$ using electricity.
    Answer: ______________________
29. The reason iron replaces copper in the reaction $\mathbf{Fe + CuSO_4 \rightarrow FeSO_4 + Cu}$.
    Answer: ______________________
30. The main environmental effects of uncontrolled combustion.
    Answer: ______________________

Answer Key

1. Combination (Synthesis) reaction
2. Decomposition reaction
3. Combustion
4. Neutralization (special double displacement)
5. Single displacement (Single replacement) reaction
6. Double displacement (Double replacement) reaction
7. Combination (Synthesis) reaction
8. Decomposition reaction
9. Photosynthesis
10. Cellular respiration
11. Decomposition reaction
12. Oxidation (Corrosion/Rusting)
13. Precipitation reaction
14. Single displacement reaction
15. Cellular respiration
16. Single displacement reaction
17. Double displacement with precipitation
18. Combustion of methane
19. Acid rain
20. Corrosion (Slow oxidation/Rusting)
21. Photosynthesis
22. Acid-carbonate double displacement reaction

23. Carbon dioxide (CO_2)
24. Photosynthesis
25. Decomposition reaction
26. Combination and oxidation reaction
27. Combustion
28. Electrolytic decomposition
29. Iron is more reactive than copper (activity series rule).
30. Global warming, acid rain, and air pollution

PART 4 – CHEMICAL EQUATIONS

Identification

Directions: Identify the term, concept, symbol, formula, or law described in each statement. Write your answer on the blank.

1. The substances found on the left side of a chemical equation.
   Answer: ______________________
2. The substances found on the right side of a chemical equation.
   Answer: ______________________
3. The meaning of the symbol **(aq)** in a chemical equation.
   Answer: ______________________
4. The meaning of the symbol **(s)** in a chemical equation.
   Answer: ______________________
5. The meaning of the symbol **(g)** in a chemical equation.
   Answer: ______________________
6. The scientific law stating that the total mass of reactants equals the total mass of products.
   Answer: ______________________
7. The meaning of the arrow (→) in a chemical equation.
   Answer: ______________________
8. The meaning of the plus sign (+) between reactants.
   Answer: ______________________
9. A balanced chemical equation has __________ atoms of each element on both sides.
   Answer: ______________________
10. Write the balanced chemical equation for the reaction of hydrogen and oxygen to form water.
    Answer: ______________________
11. The meaning of the symbol **(l)** in a chemical equation.
    Answer: ______________________
12. A number written in front of a chemical formula that indicates the number of molecules or moles.
    Answer: ______________________

13. A small number written below and to the right of an element symbol showing the number of atoms in one molecule.
    Answer: ______________________
14. The correct chemical formula for carbon dioxide.
    Answer: ______________________
15. In the formula **$CaCl_2$**, the subscript "2" refers to the number of __________ atoms.
    Answer: ______________________
16. The only numbers that may be changed when balancing a chemical equation.
    Answer: ______________________
17. In **$3H_2O$**, identify the large number 3 and the small number 2.
    Answer: ______________________
18. The balanced equation among the following: **$Ca + Cl_2 \rightarrow CaCl_2$**. What is its condition?
    Answer: ______________________
19. The total number of hydrogen atoms in **$2H_2SO_4$**.
    Answer: ______________________
20. The total number of oxygen atoms in **$4Al_2O_3$**.
    Answer: ______________________
21. In the equation **$Mg + 2HCl \rightarrow MgCl_2 + H_2$**, how many chlorine atoms are present in the product?
    Answer: ______________________
22. The first step in balancing any chemical equation.
    Answer: ______________________
23. The reason chemical equations must be balanced.
    Answer: ______________________
24. If gas escapes from an open container during a reaction, what law still applies?
    Answer: ______________________
25. In the equation **$N_2 + H_2 \rightarrow NH_3$**, the elements that are initially unbalanced are __________.
    Answer: ______________________
26. The total number of carbon atoms in **$5C_6H_{12}O_6$**.
    Answer: ______________________
27. The action that is **not allowed** when balancing chemical equations.
    Answer: ______________________
28. Balance the equation: **__ H_2 + __ $O_2 \rightarrow$ __ H_2O**.
    Answer: ______________________
29. Balance the equation: **__ N_2 + __ $H_2 \rightarrow$ __ NH_3**.
    Answer: ______________________
30. Balance the equation: **__ Al + __ $O_2 \rightarrow$ __ Al_2O_3**.
    Answer: ______________________

Answer Key

1. Reactants

2. Products
3. Aqueous (dissolved in water)
4. Solid
5. Gas
6. Law of Conservation of Mass
7. Yields / Produces / Forms
8. Reacts with (or "and")
9. Equal
10. **$2H_2 + O_2 \rightarrow 2H_2O$**
11. Liquid
12. Coefficient
13. Subscript
14. CO_2
15. Chlorine
16. Coefficients
17. **3 = Coefficient; 2 = Subscript**
18. Already balanced
19. 4 hydrogen atoms
20. 12 oxygen atoms
21. 2 chlorine atoms
22. Count the atoms of each element on both sides.
23. To obey the Law of Conservation of Mass.
24. Law of Conservation of Mass
25. Nitrogen and Hydrogen
26. 30 carbon atoms
27. Changing the subscripts
28. **2, 1, 2**
29. **1, 3, 2**
30. **4, 3, 2**

PART 5 – RATES OF CHEMICAL REACTIONS

Identification

Directions: Identify the term, concept, factor, or process described in each statement. Write your answer on the blank.

1. The measure of how fast a chemical reaction occurs.
   Answer: ______________________
2. The factor that generally increases reaction rate by increasing particle movement.
   Answer: ______________________
3. A substance that speeds up a reaction without being consumed.
   Answer: ______________________

4. A reaction that releases heat to its surroundings.
   Answer: ______________________
5. A reaction that absorbs heat from its surroundings.
   Answer: ______________________
6. The factor increased when a solid reactant is crushed into smaller pieces.
   Answer: ______________________
7. The three components of the fire triangle.
   Answer: ______________________
8. The factor responsible for slowing food spoilage inside a refrigerator.
   Answer: ______________________
9. A substance that slows down a chemical reaction.
   Answer: ______________________
10. A common example of an exothermic reaction.
    Answer: ______________________
11. The minimum amount of energy required for a reaction to begin.
    Answer: ______________________
12. The theory stating that particles must collide with enough energy and proper orientation for a reaction to occur.
    Answer: ______________________
13. The effect of crushing a solid reactant on its reaction rate.
    Answer: ______________________
14. The effect of lowering temperature on most chemical reactions.
    Answer: ______________________
15. Powdered calcium carbonate reacts faster than marble chips because it has greater __________.
    Answer: ______________________
16. Concentrated hydrochloric acid reacts faster than dilute hydrochloric acid because of higher __________.
    Answer: ______________________
17. Biological catalysts found in living organisms.
    Answer: ______________________
18. Water extinguishes fire mainly by removing __________.
    Answer: ______________________
19. A fire blanket extinguishes fire by removing __________.
    Answer: ______________________
20. Painting or galvanizing iron prevents rust by blocking contact with __________ and __________.
    Answer: ______________________
21. The food preservation method that removes water needed by microorganisms.
    Answer: ______________________
22. The food preservation method that kills microorganisms with heat and seals out oxygen.
    Answer: ______________________
23. Stirring reactants usually has what effect on reaction rate?
    Answer: ______________________
24. A process that continuously absorbs energy from sunlight.
    Answer: ______________________

25. The way a catalyst speeds up a reaction.
    Answer: ______________________
26. The reason higher temperatures greatly increase reaction rate.
    Answer: ______________________
27. The reason explosions occur extremely rapidly.
    Answer: ______________________
28. Why does a reaction eventually stop after all reactants are used up?
    Answer: ______________________
29. The device in vehicles that uses catalysts to reduce harmful exhaust gases.
    Answer: ______________________
30. The factor increased when oxygen concentration becomes higher, causing fire to burn faster.
    Answer: ______________________

Answer Key

1. Reaction rate
2. Temperature
3. Catalyst
4. Exothermic reaction
5. Endothermic reaction
6. Surface area
7. Fuel, oxygen, and heat (ignition temperature)
8. Lower temperature
9. Inhibitor
10. Combustion (burning wood)
11. Activation energy
12. Collision theory
13. Increases the reaction rate
14. Slows the reaction rate
15. Surface area
16. Concentration
17. Enzymes
18. Heat
19. Oxygen
20. Oxygen and water (moisture)
21. Drying, salting, or sugaring
22. Canning
23. Speeds up the reaction
24. Photosynthesis
25. By lowering the activation energy
26. More frequent collisions and more particles have enough energy to react
27. Instant gas release, heat, and large surface contact

28. The reactants are completely used up, so no more effective collisions occur.
29. Catalytic converter
30. Concentration of oxygen (reactant concentration)

PART 6 – HOMEOSTASIS

Identification

Directions: Identify the term, concept, organ, hormone, or process described in each statement. Write your answer on the blank.

1. The process of maintaining a stable internal environment despite external changes.
   Answer: ______________________
2. The two body systems that serve as the main control systems for homeostasis.
   Answer: ______________________
3. The type of feedback mechanism that reverses a change and returns the body to its normal condition.
   Answer: ______________________
4. The normal or target value maintained by the body during homeostasis.
   Answer: ______________________
5. The structure that detects changes in the internal or external environment.
   Answer: ______________________
6. The structure that carries out the response to restore balance.
   Answer: ______________________
7. The main function of negative feedback in the body.
   Answer: ______________________
8. The body characteristic that is **not** maintained by homeostasis according to the lesson.
   Answer: ______________________
9. The part of the brain responsible for regulating body temperature.
   Answer: ______________________
10. The response of blood vessels in the skin when the body is cold.
    Answer: ______________________
11. The body system that uses hormones as chemical messengers.
    Answer: ______________________
12. The two hormones responsible for regulating blood glucose levels.
    Answer: ______________________
13. The hormone released by the pancreas when blood sugar becomes too high.
    Answer: ______________________
14. The hormone released by the pancreas when blood sugar becomes too low.
    Answer: ______________________
15. The organ that regulates water and salt balance by adjusting urine concentration.
    Answer: ______________________
16. The body's response that generates heat when a person feels cold.
    Answer: ______________________

17. The type of feedback mechanism demonstrated by shivering.
    Answer: ______________________
18. The relationship between insulin and glucagon in regulating blood sugar.
    Answer: ______________________
19. The type of feedback mechanism that amplifies a change away from the set point.
    Answer: ______________________
20. The hormone released after eating to lower blood glucose levels.
    Answer: ______________________
21. The organ that mainly removes waste while helping regulate body fluids.
    Answer: ______________________
22. The two body responses that help lower body temperature when a person becomes too hot.
    Answer: ______________________
23. A common example of positive feedback in the human body.
    Answer: ______________________
24. Homeostasis requires a constant supply of what to maintain balance?
    Answer: ______________________
25. If blood pH is not properly maintained, what happens first to many enzymes?
    Answer: ______________________
26. During exercise, breathing rate increases to maintain the balance of which two gases?
    Answer: ______________________
27. Arrange the correct sequence of a homeostatic feedback loop.
    Answer: ______________________
28. The hormone that strengthens uterine contractions during childbirth through positive feedback.
    Answer: ______________________
29. The long-term result if homeostasis fails to maintain stable internal conditions.
    Answer: ______________________
30. The active process that continuously keeps the body's internal environment stable.
    Answer: ______________________

Answer Key

1. Homeostasis
2. Nervous system and Endocrine system
3. Negative feedback
4. Set point
5. Receptor (Sensor)
6. Effector
7. Reverse the change and restore the set point
8. Constant body height
9. Hypothalamus
10. Vasoconstriction (blood vessels constrict)

11. Endocrine system
12. Insulin and Glucagon
13. Insulin
14. Glucagon
15. Kidneys
16. Shivering
17. Negative feedback
18. Opposing negative feedback
19. Positive feedback
20. Insulin
21. Kidneys
22. Sweating and vasodilation
23. Blood clotting
24. Energy
25. They denature or stop functioning properly.
26. Oxygen (O_2) and Carbon dioxide (CO_2)
27. Stimulus → Receptor → Control Center → Effector → Response
28. Oxytocin
29. Disease, organ damage, or death
30. Homeostasis

PART 7 – MECHANISMS OF EVOLUTION

Identification

Directions: Identify the term, concept, scientist, process, or evidence described in each statement. Write your answer on the blank.

1. The gradual change in the inherited traits of a population over many generations.
   Answer: ______________________
2. The scientist who proposed natural selection as the primary mechanism of evolution.
   Answer: ______________________
3. The process in which organisms best adapted to their environment survive and reproduce.
   Answer: ______________________
4. An inherited characteristic that helps an organism survive and reproduce in its environment.
   Answer: ______________________
5. The natural differences in traits among individuals of the same species.
   Answer: ______________________
6. The passing of genetic traits from parents to offspring.
   Answer: ______________________
7. Preserved remains, traces, or impressions of ancient organisms found in rocks.
   Answer: ______________________

8. Body parts with similar internal structures but different functions due to common ancestry.
   Answer: ______________________
9. A reduced structure that has lost most or all of its original function through evolution.
   Answer: ______________________
10. The study of the geographic distribution of living organisms.
    Answer: ______________________
11. The famous example of evolution involving light and dark peppered moths during the Industrial Revolution.
    Answer: ______________________
12. The type of isolation that occurs when a physical barrier separates populations.
    Answer: ______________________
13. Structures that perform the same function but have different evolutionary origins.
    Answer: ______________________
14. The process that prevents gene flow between separated populations, leading to the formation of new species.
    Answer: ______________________
15. Fossils found in deeper rock layers are generally __________ than those in younger layers.
    Answer: ______________________
16. The evolutionary explanation for the long necks of giraffes.
    Answer: ______________________
17. The process in which humans choose organisms with desirable traits for breeding.
    Answer: ______________________
18. The study of similarities in the early development of embryos as evidence for evolution.
    Answer: ______________________
19. The type of isolation that occurs when populations do not interbreed because of differences in behavior or breeding time.
    Answer: ______________________
20. A species found naturally in only one geographic location.
    Answer: ______________________
21. The process through which new species are formed.
    Answer: ______________________
22. The primary mechanism of evolution that consistently produces adaptations suited to the environment.
    Answer: ______________________
23. Tiny unused pelvic bones in modern whales are an example of what type of evolutionary evidence?
    Answer: ______________________
24. In science, a well-tested explanation supported by extensive evidence is called a __________.
    Answer: ______________________
25. The supercontinent whose existence is supported by matching fossils found in South America and Africa.
    Answer: ______________________

26. A condition required for natural selection in which inherited differences already exist within a population.
    Answer: ______________________
27. Homologous structures provide evidence that different species share a __________.
    Answer: ______________________
28. During a long drought, finches with larger beaks survived better because of __________.
    Answer: ______________________
29. The movement of individuals between populations that keeps their gene pools similar.
    Answer: ______________________
30. The strongest evidence for evolution comes from multiple independent lines of evidence, including fossils, homologous structures, DNA similarity, and __________.
    Answer: ______________________

Answer Key

1. Evolution
2. Charles Darwin
3. Natural selection
4. Adaptation
5. Variation
6. Heredity (Inheritance)
7. Fossils
8. Homologous structures
9. Vestigial structure
10. Biogeography
11. Natural selection (Peppered moth example)
12. Geographic isolation
13. Analogous structures
14. Reproductive isolation
15. Simpler and more different from modern organisms
16. Natural selection favored longer-necked giraffes over many generations.
17. Artificial selection
18. Comparative embryology
19. Reproductive isolation
20. Endemic species
21. Speciation
22. Natural selection
23. Vestigial structure
24. Scientific theory
25. Pangaea
26. Variation
27. Common ancestor
28. Natural selection

29. Gene flow
30. Biogeography